

1. Count the following collections:

- The number of integers $k \in \{1, 2, \dots, 100\}$ so that $2 \mid k$ or $5 \mid k$.
- The number of three-letter strings (e.g. ABC, XYZ).
- The number of ways of splitting a group of 10 people into two groups of five.

Solution:

- Use the subtraction rule. Let A be integers divisible by 2 and B integers divisible by 5. Then $|A| = 50$ and $|B| = 20$. Note $A \cap B$ consists of integers divisible by 10, so $|A \cap B| = 10$. Hence:

$$|A \cup B| = |A| + |B| - |A \cap B| = 50 + 20 - 10 = 60$$

- There are 26 letters, so by the product rule the number of three letter strings is 26^3 .
- We just must choose one group of size 5 from 10, which we can do in $\binom{10}{5}$ ways (the unchosen people form the second group).

2. Count strings of three letters followed by three digits with no repeated characters (e.g. $ABC123$)?

Solution: By the product rule, the number of three letter strings with no repeated characters is $26 \cdot 25 \cdot 24$. Likewise, the number of 3 digit strings with no repeated digits is $10 \cdot 9 \cdot 8$. Hence in total we get:

$$26 \cdot 25 \cdot 24 \cdot 10 \cdot 9 \cdot 8$$

3. How many 5 letter strings are there with at least one vowel?

Solution: It's easier to count the number of strings *not* satisfying the desired property, and subtract it from the total. Namely, since there are five vowels, there are 21^5 five-letter strings with no vowels. In total, there are 26^5 five-letter strings. Hence the number of five letter strings with at least one vowel is:

$$26^5 - 21^5$$

4. Show that the number a_n of subsets of $\{1, 2, \dots, n\}$ that do not contain consecutive integers equals the Fibonacci number f_{n+2} . Recall that f_n is defined via $f_n = f_{n-1} + f_{n-2}$ with $f_1 = f_2 = 1$.

Solution: We prove this by induction. For our base case, note that when $n = 0$ we have a single subset $\emptyset$, so $a_0 = 1$. When $n = 1$, we have the two subsets $\emptyset, \{1\} \subseteq \{1\}$, so $a_1 = 2$. These equal f_2, f_3 respectively.

Suppose then for induction that $a_k = f_k$ for $k < n$. We aim to show $a_n = f_n$. Consider a subset $S \subseteq \{1, \dots, n\}$ that contains no consecutive integers. If $n \in S$, then we know $n-1 \notin S$. Moreover, $S \cap \{1, \dots, n-2\}$ is again a subset of $\{1, \dots, n-2\}$ with no consecutive integers. If $n \notin S$, then $S \cap \{1, \dots, n-1\}$ is a subset of $\{1, \dots, n-1\}$ with no consecutive integers. Moreover, clearly *any* subset of $\{1, \dots, n-1\}$ or $\{1, \dots, n-2\}$ with no consecutive integers can appear in this way. Hence:

$$a_n = a_{n-1} + a_{n-2} = f_{n-1} + f_{n-2} = f_n$$

as desired.

5. For fixed $1 \leq k \leq n$, how many distinct cycles of length k are there in the complete graph K_n ?

Solution: Start by choosing which k vertices make up the cycle, which we can do in $\binom{n}{k}$ ways. Fix one vertex arbitrarily to be the starting vertex. Then we can decide the order of the remaining $k-1$ vertices in $(k-1)!$ ways. This over-counts by a factor of two, since reversing the order produces the same cycle (e.g. the cycle $v_1 \rightarrow v_2 \rightarrow v_3 \rightarrow v_1$ is the same as the cycle $v_1 \rightarrow v_3 \rightarrow v_2 \rightarrow v_1$ just read in reverse). Hence in total we have:

$$\binom{n}{k} \cdot \frac{(k-1)!}{2}$$

6. Define a simple walk of length $2n$ as a sequence of integers $(x_0, x_1, x_2, \dots, x_{2n})$ so that $(x_{i+1} - x_i) \in \{-1, 1\}$ for all i . For example, $(0, 1, 2, 1, 0)$ is a simple walk of length four.

- (i) How many simple walks are there of length $2n$ that end at zero ($x_{2n} = 0$)?
(ii) How many simple walks are there of length $2n$ that end at zero and are always nonnegative? Hint¹

Solution:

- (i) In order to start and end at zero, we must choose exactly n "up" steps ($x_{i+1} - x_i = 1$) and n "down" steps ($x_{i+1} - x_i = -1$). There are $\binom{2n}{n}$ ways to choose the location of the n up steps among the $2n$ steps.
(ii) We know there are $\binom{2n}{n}$ walks total, so we will instead count the exceptional set: walks which are negative at some point in time. Suppose that (x_i) is one such walk. Let:

$$j := \inf\{i : x_i = -1\}$$

be the first time the walk becomes negative. Define a new walk (y_i) via:

$$y_i = x_i \text{ for } i \leq j, \quad y_{i+1} - y_i = -(x_{i+1} - x_i) \quad \text{for } i \geq j.$$

¹count the number of walks not satisfying this property. Try to relate this count to walks from 0 to -2 in $2n$ steps

Namely, the walk (y_i) follows the same path as the walk (x_i) until the first time the latter becomes negative. After that point, it reflects the walk: every time (x_i) goes up we have (y_i) go down, and vice versa. We claim this reflection operation $(x_i) \mapsto (y_i)$ gives a bijection from the set of walks from 0 to 0 that are negative at some point to the set of walks from 0 to -2 . The latter set has size $\binom{2n}{n-1}$ by the same argument as in problem (i), so the bijection implies the former set has the same size. Hence the number of walks of length $2n$ from 0 to 0 and are always nonnegative has size:

$$\binom{2n}{n} - \binom{2n}{n-1}$$